GEONHO KIM

Bachelor Student

Chungbuk University

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Education

Mar. 2021 ~ Present

Chungbuk National University

Major - 지능로봇공학과 (Dept. of Intelligent Systems & Robotics)

Minor : Autonomous Driving

Bachelor Student

GPA: 3.61 / 4.5

Cheongju,
Korea

Teaching Experience

Digital System Design Undergraduate Teaching Assistant (Advisor: Prof. Chansik Park) [*09. 2025 – 12. 2025]*

- Co-managed laboratory sessions as one of two Tas
- Instructed students on designing digital logic circuits using **FPGA** boards and **Xilinx** tools.
- Assisted students in debugging hardware implementation issues and provided technical guidance during practice hours.

Leadership & Activities

Clothoid-R [*01. 2025 – Present] Team Member / Team Leader*

- Leading the autonomous driving research team 'Clothoid-R'.
- Directing overall project schedules and managing team resources for upcoming competitions and research projects.

2025 Future Vehicle Global Leadership Program [*01. 2026] Participant*

- Selected for the university-sponsored global field trip program focused on future mobility trends.

- Visited Automotive World 2026 in Tokyo, Japan, to research the latest industrial technologies in autonomous driving and EVs.

Research Interests

- Self driving
- GPS
- Localization system

Conferences

1. KIM GEONHO, PARK CHANSIK, " **Comparison of Accuracy and Implementation Feasibility of SPP, PPP, and RTK Techniques for Autonomous Vehicle Applications Using RTKLIB**", IPNT, Jeju, Korea (Nov. 2025) – Oral

Research & Professional Experiences

- Research Student at Department of Intelligent Systems and Robotics, SANS Lab, Korea (Feb. 2025 ~ Present)
- **Partridge Systems** (Seoul) Intern [06. 2026 - Present]

Awards and Honors

- Advanced 자율주행 로봇 레이스 1/2 최우수상, 국제e-모빌리티엑스포, Korea (Jul. 2025)

Project experience

Intelligent Posture Correction Car Seat (Capstone Design)

Leader / Main Developer [09. 2025 - 06. 2026]

- Developed an intelligent posture correction car seat utilizing sensor fusion of vision-based pose estimation and pressure data to physically adjust user seating.
- Integrated MediaPipe Pose to calculate vertical neck angles and built a real-time calibration system using multi-channel FSR sensors via Arduino.
- Implemented multi-threading for synchronized camera data processing and designed sensor-fusion logic to trigger discrete vehicle seat motor controls (Slide, Tilt, Lumbar, Lift) via serial communication.

KSAE 2025 Autonomous Formula Student Competition (Simulation) (Busan)

Localization Engineer / Team Member [06. 2025] - [11. 2025]

- Organized by the Korean Society of Automotive Engineers (KSAE).
- Responsible for the Localization module in a high-speed autonomous racing simulation.
- Developed and implemented an Extended Kalman Filter (EKF) algorithm to improve the accuracy of vehicle state estimation and position tracking.

Seoul Residential Satisfaction Prediction and Factor Analysis (Python-based) (Chungbuk National University)

Personal [03. 2025 - 06. 2025]

- Term project for the Artificial Intelligence course.
- Developed machine learning models utilizing Multi-Layer Perceptron and Decision Tree algorithms to predict the residential satisfaction of citizens in Seoul
- Investigated and identified the key factors that most significantly influence housing satisfaction.

Autonomous Driving Simulation with We-bot (Chungbuk National University)

Leader [09. 2025] - [12. 2025]

- Term project for the Robot Programming course.
- Implemented a vision-based autonomous driving system using the We-bot platform and camera sensors.
- Developed line-tracking algorithms to recognize road lanes and control the robot's steering and velocity in real-time.

4th International University Student EV Autonomous Driving Competition (Jeju, S.Korea)

Localization Engineer / Team Member [03. 2025] - [07. 2025]

- Hosted by the International Electric Vehicle Expo (IEVE); a competitive race involving 1/2 scale electric vehicles.
- Led the development of the Localization system for the actual hardware vehicle platform.
- Focused on resolving GPS signal instability

Smart Safe with Biometric Security (S.Korea)

Leader [03. 2025] - [06. 2025]

- Term project for the Microprocessor course.
- Built a hardware-based security safe integrating Facial Recognition and GPS modules.
- Implemented an alert system that pushes real-time notifications and GPS coordinates to a smartphone upon unauthorized access attempts or security breaches.

Algorithmic Music Composition Program (C-based) (Chungbuk National University)
Personal [*03. 2024] - [06. 2024]*

- Term project for the Open-Source Project 1 course.
- Developed an automatic music generation program using C language.
- Designed algorithms to structure musical notes and rhythms, creating unique compositions without human intervention.

Advanced Music Synthesis & Composition System (C++) (Chungbuk National University)

Personal [*09. 2024] – [12. 2024]*

- Term project for the Open-Source Project 2 course.
- Refactored and expanded the previous composition engine using C++ to enhance modularity and performance.
- Integrated audio signal processing to map generated notes to specific instrument sounds, enabling realistic audio playback of the composed pieces.

Skills and Techniques

- Python
- C++
- RTKLIB
- ROS

Relevant Coursework

- **Robotics & Control:** Automatic Control, Modern Control, Robot Programming, Artificial Intelligence, Digital Signal Processing, Sensor Engineering
- **Mathematics & Signal:** Linear Algebra, Probability and Statistics, Signal and System, Engineering Mathematics, Calculus
- **Computer Science & Embedded:** Data Structure and Algorithm, Operating System, Microprocessor, Object Oriented Programming, Digital System Design